Tutorial 3.2

a) What is polarization? How polarization can create a channel in a E-MOSFET?

b) What is fixed/self /voltage divider biasing? Draw the load line and Q point of a fixed biased JFET amplifier. How you can adjust the Q point of the fixed biased JFET amplifier?

c) Draw a small signal AC model for an N-channel JFET in CS connection and explain all the values.

d) Find out the voltage gain expression of a fixed/self/voltage divider biased JFET amplifier in CS connection.

Problem 1:

a) What is the total output voltage for the unloaded amplifier in Figure 4? I_{DSS} is 4.3 mA and $V_{GS(off)}$ is -2.7 V. If a 4.7 kΩ resistor is AC coupled to the output of the amplifier below, what is the resulting output voltage?

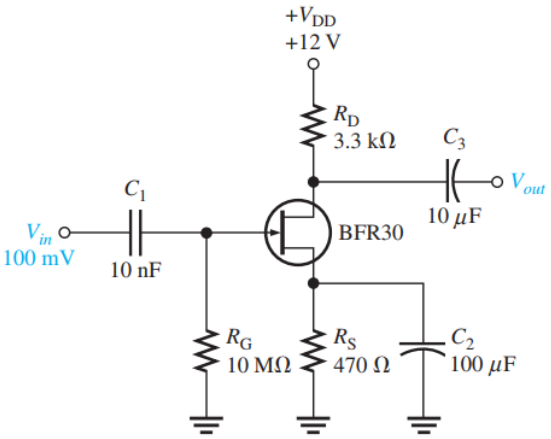


Figure 4

b) What is the input resistance seen by the signal source in the circuit in Figure 5? I_{GSS} = 30 nA at V_{GS} = 10 V.

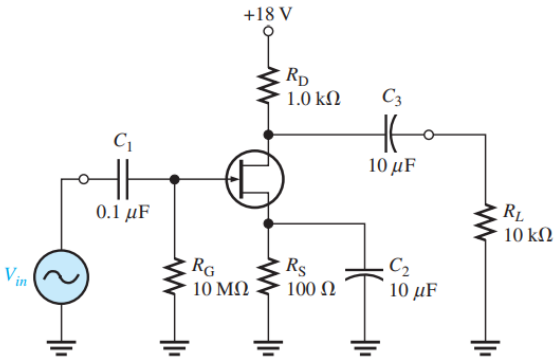


Figure 5

Problem 2

Find V_{GS}, I_D, V_{DS} and the AC output voltage for the circuit in Figure 6. Assume that $I_{D(on)}$ = 200 mA at V_{GS} = 4 V, $V_{GS(th)}$ = 2 V, g_m = 23 mS. V_{in} = 25 mV.

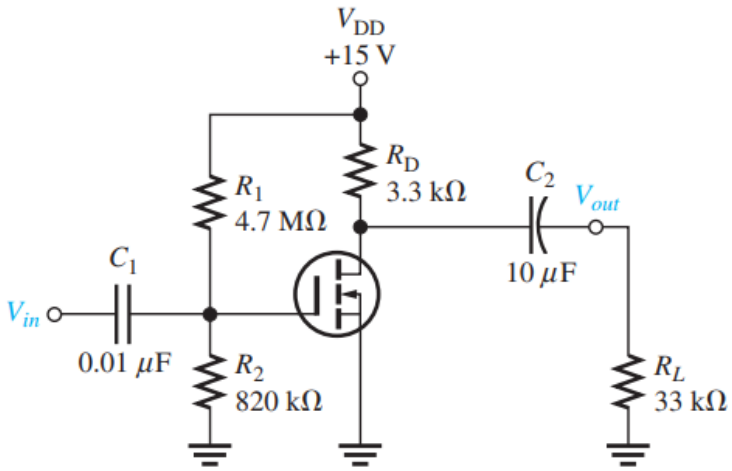


Figure 6